

Vercel Solutions Architect Take-Home Assessment

Meridian Capital Executive Summary

Meridian Capital is growing faster than its IT organization can ship: a ticket-gated, four-environment release process turns a same-day idea into a multi-week deployment, and that gap is starting to constrain the feature velocity the business depends on to meet its customers needs.

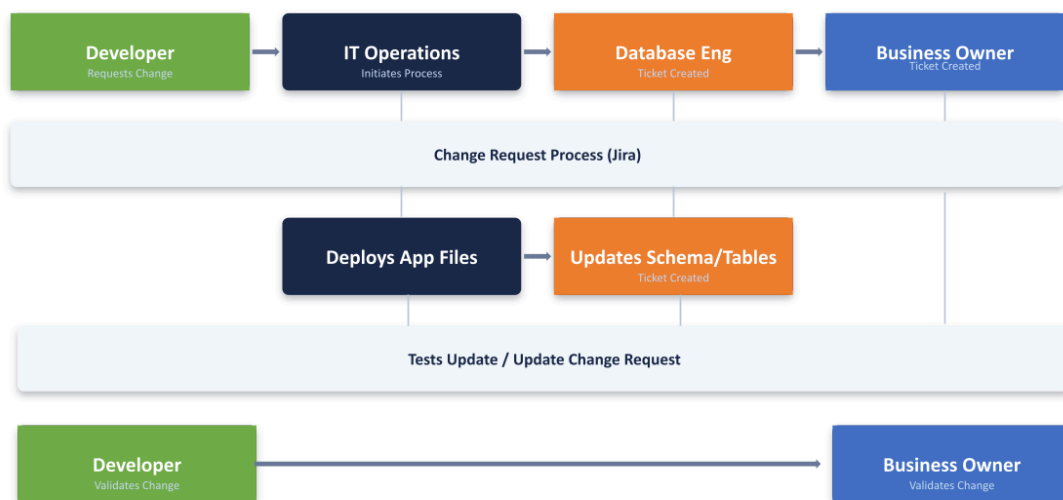
Closing the gap between an idea and a deployed feature is what will let Meridian's developers keep shipping for its customers at the pace the business is growing. Vercel can close that gap.

In the current architecture (see below), the front-end, middleware, and database tiers behind Meridian's Portfolio Watchlist app are each owned by a different team; Enterprise IT runs the AWS-hosted front-end and middleware, Enterprise DB runs the on-premises database so a single deployment to Dev, QA, or Staging means coordinating across all of them as well as the business owner and developers. A routine promotion can require:

- Provisioning new systems
- Updating database schemas and tables
- Applying security patches
- Clearing caches and diagnosing network issues

CURRENT STATE

Every Environment, Same Bottleneck



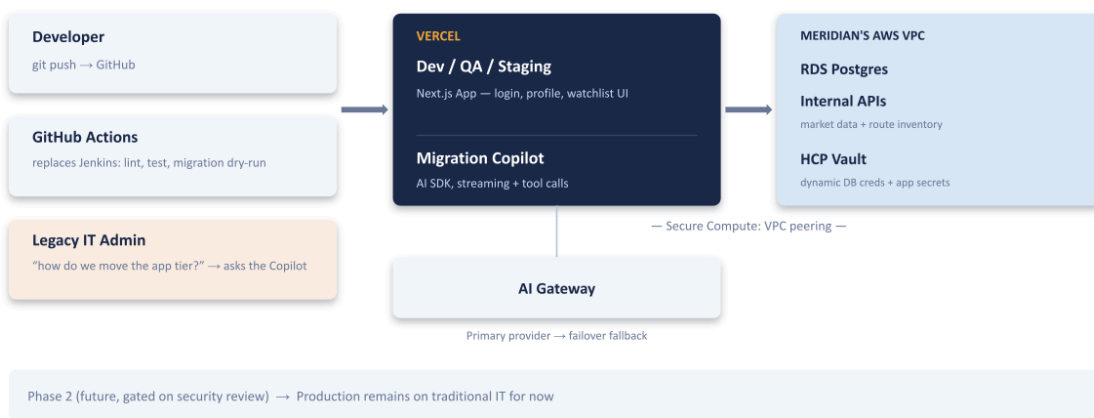
At Meridian today, an idea waits in a ticket queue. On Vercel, it's a preview URL before the meeting about it even ends. Today, Meridian's front-end and middleware run in AWS under Enterprise IT, the database runs on-prem under Enterprise DB, and every change to any of them goes through a ticket queue and a manual, one-environment-at-a-time CI pipeline.

With Vercel Dev/QA/Staging move to git-push Preview Deployments with instant rollback, feature flags, preview deployments and real-time feedback.

TARGET STATE

Preview Deployments + Secure Compute

Production stays on traditional IT for Phase 1 — Dev/QA/Staging move first



The overview of the solution Vercel is proposing can be found [here](#). This will gain Meridian Capital the following;

- **Preview Deployments + Rolling Releases (Replace).** Every PR gets an isolated preview URL and QA/Staging promote via percentage-based rollout with instant rollback, replacing ticket-gated environments and manual Jenkins promotion.
- **AI SDK + AI Gateway (Augment).** Powers the Migration Copilot with one API across providers, automatic failover, and a cost/token dashboard Compliance can review.
- **Secure Compute + OIDC Federation (Connect).** A private network path into Meridian's AWS VPC, using short-lived Vercel-issued tokens instead of standing secrets.
Maintaining the high level of federation and security that Meridian requires.

Live demo: <https://meridian-migration-demo.vercel.app>

Repo: <https://github.com/FraserPol/meridian-migration-demo>

Success Criteria

Baseline → Target

	BASELINE		TARGET
Environment provisioning	Ticket queue (days)	→	Preview URL on git push (minutes)
Secrets	Long-lived, manually rotated	→	Dynamic, TTL-bound — zero static DB secrets
Rollback	Manual redeploy (~30–60 min)	→	Instant rollback to prior deployment (seconds)
AI spend visibility	None	→	Per-request / per-feature cost + token dashboard
Migration guidance	Tribal-knowledge cutover planning	→	AI-generated, reviewable rewrite/rollback plan

Iterating To Greatness - Next Steps

The recommended path forward is phased and evidence-based.

- **Phase One - Provision Development on Vercel**
 - Validates the OIDC/Secure Compute pattern under a sustained soak
 - Ensure that regulatory, security and compliance requirements are all still met
 - Work directly with Meridian's security team to close the items a production trust-boundary review will require;
 - A defined PII and data-residency posture for any prompt content leaving the boundary
 - An audit trail integrated with Meridian's SIEM via Log Drains
 - An RBAC model governing who may promote or roll back a deployment
 - All relevant SOC 2 Type II, ISO 27001, and DPA documentation
- **Phase Two - Provision QA and Staging to Vercel**
 - Review measured results from Development and a completed security review
- **Phase Three - Review Production Requirements**
 - Determine if Production can be migrated to Vercel (currently outside of scope)